

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Tile – Travel Planner Agent

Domain of Project – Travel and Tourism

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement -

The Challenge : Planning a trip is often stressful and time-consuming. Travelers have to juggle scattered information across dozens of websites to find flights, hotels, and local activities that fit their specific budget and dates. Unexpected weather or sudden cancellations can quickly ruin carefully made plans. Without a single, intelligent assistant to organize, monitor, and adapt these details in real time, travel planning remains a complex and frustrating chore rather than an enjoyable experience.

The Objective : Build an AI-powered Travel Planner Agent that acts as a smart, 24/7 personal travel assistant. The solution should:

- **Personalized Itinerary Building:** Create custom daily travel plans based on user budgets, travel dates, personal interests, and physical constraints.
- **Real-Time Data Retrieval:** Fetch live weather updates, map routes, and local guide information to keep the travel plan accurate and safe.
- **Booking and Logistics Support:** Recommend the best transport options and accommodations, helping users manage their reservations in one place.
- **Dynamic Trip Optimization:** Alert users to sudden changes like flight delays or bad weather and instantly suggest backup plans to keep the trip moving smoothly.

Proposed Solution -

The proposed solution is an AI-powered Travel Planner Agent built using IBM watsonx Orchestrate and IBM Granite models to provide personalized, conversational travel planning. The system addresses challenges in trip organization, budgeting, and itinerary creation through an intelligent, Q&A-based chat interface.

At its core, the solution relies on advanced prompt engineering and strict behavioral configurations rather than complex backend databases. By defining precise system rules, conversational boundaries, and user personas, the agent leverages the reasoning capabilities of IBM Granite models to generate custom travel itineraries and contextual recommendations strictly through natural language interactions.

The system operates as a unified, text-based conversational assistant capable of handling multiple travel planning aspects dynamically:

- **Personalized Itinerary Generation:** Engages users in a fluid Q&A format to gather travel dates, budgets, and preferences, instantly generating detailed day-by-day travel plans.
- **Logistics & Budget Guidance:** Suggests suitable transport options and accommodation styles, ensuring all conversational recommendations align with the user's stated financial constraints.
- **Dynamic Trip Adjustment:** Allows users to input situational changes (e.g., "The forecast shows rain tomorrow") and instantly responds with optimized backup schedules and indoor activities.
- **Destination Discovery:** Provides tailored recommendations for local attractions, historical sites, or dining options based on the user's conversational inputs.

Technology Used

- **IBM watsonx Orchestrate**: For defining the agent's behavior, setting conversational boundaries, and managing the chat interface visually without writing backend code.
- **IBM Cloud Lite**: Provides the free, reliable infrastructure needed to run the watsonx services securely and host the agent's brain.
- **IBM watsonx.ai**: For accessing the Granite models, managing the underlying prompt instructions, and powering the conversational logic.
- **HTML, CSS, and Vanilla JavaScript**: For designing and building the clean, responsive front-end landing page that serves as the user interface.
- **GitHub Pages** (<https://krishujeniya.github.io/Wanderlust/>): For hosting and deploying the final static website, integrating the watsonx embed script, and making the Travel Planner Agent live and accessible on the internet.

Orchestrate component Used -

1. **Chat Input:** Interface where users type their travel queries, dates, budget constraints, and destination preferences directly into the web chat.
2. **IBM watsonx Orchestrate AI Agent:** Processes user input and applies your custom travel rules, using Granite models to generate personalized itineraries and handle travel logistics.
3. **Name of IBM Granite model:** gpt-oss-120b(due to default). This model is highly effective for complex conversational reasoning, constraint management, and formatting detailed daily schedules.
4. **Chat output:** Displays the AI-generated responses, such as custom day-by-day travel plans, alternative route suggestions, and budget breakdowns.
5. **System Behavior Configuration:** Since your project relies on agent behavior rather than external databases, this acts as the core memory component. It stores the strict system prompts, conversational boundaries, and persona rules directly inside watsonx Orchestrate.
6. **GitHub Pages Web Integration:** Your custom frontend static website built with HTML, CSS, and JavaScript, which uses the IBM embed script to host and display the chat interface to the public.

Agent Instructions Screenshot-

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

Home / Manage agents / Travel Planner Agent

Test agentDeploy

Travel Planner Agent

Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Toolset

Add tools and other agents to help your agent take action.

Behavior

Define how and where your agent should react to requests and respond to users.

Instructions

You are a friendly, helpful, and highly organized travel guide who speaks in clear, simple English and sounds like a helpful friend. Always start the conversation by saying, "Hello! I am your personal Travel Planner. Tell me where you want to go, how many days you have, and your budget, and I will build the perfect trip for you!" Your only job is to plan travel, suggest places to visit, and manage itineraries through this text chat window, keeping your answers short and easy to read. If a user asks you to do things outside of travel planning like math or homework, politely reply that you can only help with vacations, and never promise to buy tickets for them since you only provide planning suggestions.

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add rules that guide an agent's actions and responses.

Build with AITalk to agent

Running in debug mode

Today

You 11:00 AM

Plan a 4-day trip to Goa for two people focusing on historical sites with a budget of 30,000 INR.

Travel Planner Agent 11:00 AM

Goa-Historic-Highlights 4-Day Itinerary

(Two travelers, total budget ~ ₹30,000 – includes transport, accommodation, food, entry fees & a little buffer)

QUICK OVERVIEW

Day	Main Theme	Key Sites (in logical order)	Approx. Cost (INR)
Day 1	Arrival & Old-Goa churches	Dabolim Airport → Panjim → Old-Goa (Basilica of Bom Jesus, Se Cathedral, Church of St Catherine) → Panaji promenade	7500
Day 2	Portuguese forts & forts of the South	Fort Aguada, Sinquerim Beach, Chapora Fort, Tiracol Fort (optional ferry)	7000
Day 3	Heritage villages & museums	São Tomé (Heritage House), Mona Palace, Goa State Museum, Archaeological Survey of India (ASI) Museum	6500
Day 4	Colonial architecture & departure	Latin Quarter (Fontainhas), Reverend Moses' House, Mar... Vijaykar's Heritage Walk, last-minute shopping → Airport	7000

Save as test

Type something...

Prompt with output Screenshot:

The screenshot displays the IBM watsonx Orchestrator interface. The top navigation bar includes links to various services like Kimi, MongoDB, Github, Kaggle, Upwork, Freelancer, Fiver, Sollearn, India AI, Cloud Skills, Cloud Console, Opal, Dashboard, Prompt Optimizer, GitHub Student De..., MongoDB for Stud..., and Download your Te... The main header shows 'IBM watsonx Orchestrator' with options for English, Contact Sales, Buy Now, and a 30-day trial. The breadcrumb trail indicates the current location: Home / Manage agents / Travel Planner Agent.

The interface is divided into two main sections. The left section, titled 'Travel Planner Agent', contains a sidebar with navigation links: Profile, Knowledge, Toolset, Behavior, and Channels (Preview). The main content area for the 'Profile' tab includes a description of the agent's purpose, a detailed description of its capabilities (real-time data, personalized plans, integrated maps, booking management), a 'Welcome message' field, and a 'Quick start prompts' section with pre-set messages for trip planning and homestay recommendations.

The right section shows the agent's output in 'Running in debug mode'. It displays the time '11:03 AM' and the agent's name 'Travel Planner Agent'. Below this, a disclaimer states 'Accuracy of generated answers may vary. Please double-check responses.' Three example prompts are shown in boxes, each with a right arrow: 'Plan a 4-day trip to Goa for two people focusing on...', 'Recommend the best homestays and train routes...', and 'Create a budget-friendly 3-day itinerary for a solo trip t...'. The first prompt is highlighted with a blue background.

Prompt with output Screenshot:

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

Home / Manage agents / Travel Planner Agent

Test agentDeploy

Build with AITalk to agent

Running in debug mode

TodayYou 11:10 AM

Plan a 4-day trip to Goa for two people focusing on historical sites with a budget of 30,000 INR.

Travel Planner Agent 11:10 AM

Goa Historical Highlights - 4-Day Itinerary for 2 People

Budget: ~ 30,000 INR (including accommodation, food, transport & entry fees)

Day	Morning	Mid-day	Afternoon / Evening	Budget (INR)
Day 1 - Arrival & Old Goa	• Arrive at Dabolim Airport / train station • Taxi/auto to Hotel Casa De Goa (Panaji) - budget-friendly heritage hotel (~ 2,800 ₹/night)	• Breakfast at hotel • Visit Basilica of Bom Bom (free)	• Explore Se Cathedral (₹20 entry) • Walk to Church of St. Catherine (₹20) • Sunset at Miramar Beach (free)	Hotel 2,800 + transport 400 + entry 40 = ~3,240
Day 2 - Portuguese Forts	• Breakfast at hotel • Hire a scooter (₹350/day) or auto for the day	• Ride to Fort Aguada (free, 1-hour tour)	• Lunch at a local shack (₹300 for two) • Visit Fort Tiracol (₹30 entry, ferry ₹150 round-trip) • Evening back in Panaji, dinner at Mango Tree (₹600)	Scooter 350 + lunch 300 + ferry 150 + entry 30 + dinner 600 = ~1,430
Day 3 -		• Explore Latin Quarter	• Lunch at Mum's Kitchen (₹800) • Visit Goa State Museum (₹30)	Auto 200 +

Type something...

Travel Planner Agent

AI Model: GPT-OSS 120B -- Ope...

Embed on your website

Test your agent easily. Copy and paste this snippet into your environment to preview before deploying. [Learn more](#)

```
<script>
window.wxOConfiguration = {
  orchestrationID: "9b8df9c9897a405b95d55653bb408ace_9d5fb77d-ea4e-4ba3-8969-191dc3d7116a",
  hostURL: "https://us-south.watsonx-orchestrate.cloud.ibm.com",
  rootElementID: "root",
  deploymentPlatform: "ibmcloud",
  crn: "crn:vl:bluemix:public:watsonx-orchestrate:us-south:a/9b8df9c9897a405b95d55653bb408ace",
  chatOptions: {
    agentId: "5c61d087-e566-4eb0-9a18-3429e0b0a334",
  }
};
setTimeout(function () {
  const script = document.createElement('script');
  script.src = `${window.wxOConfiguration.hostURL}/wchat/wxolloader.js?embed=true`;
  script.addEventListener('load', function () {
    wxolloader.init();
  });
  document.head.appendChild(script);
}, 0);
</script>
```

Show less

Teams

Enable your agent to communicate via Teams

WhatsApp with Twilio

Enable your agent to communicate via WhatsApp with Twilio

Facebook messenger

Prompt with output Screenshot:

IBM watsonx Orchestrator

EnglishContact SalesBuy Now30 trial days leftA? ⓘ KLU

Home / Manage agents / Travel Planner Agent

Travel Planner Agent

AI Model: GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Voice mode is not selected

Embed on your website

Test your agent easily. Copy and paste this snippet into your environment to preview before deploying.
[Learn more](#)

```
<script>
  window.wxOConfiguration = {
    orchestrationID: "9b8df9c9897a405b95d55653bb408ace_9d5fd77d-eade-4ba3-8969-191dc3d7116a",
    hostURL: "https://us-south.watsonx-orchestrator.cloud.ibm.com",
    rootElementID: "root",
    deploymentPlatform: "ibmcloud",
    crn: "crn:vl:bluemix:public:watsonx-orchestrator:us-south:a/9b8df9c9897a405b95d55653bb408ace",
    chatOptions: {
      agentId: "5c61d087-e566-4eb0-9a18-3429e0b0a334",
    }
  };
}
setTimeout(function () {
  const script = document.createElement('script');
  script.src = `${window.wxOConfiguration.hostURL}/wxchat/wxolloader.js?embed=true`;
  script.addEventListener('load', function () {
    wxolloader.init();
  });
  document.head.appendChild(script);
}, 0);
</script>
```

Show less

Teams

Enable your agent to communicate via Teams

WhatsApp with Twilio

Enable your agent to communicate via WhatsApp with Twilio

Build with AI

Talk to agent

Options

Running in debug mode

Day 4 – South Goa History & Departure	• Check-out, store luggage at hotel • Breakfast & head south by scooter	• Visit Shri Mangeshi Temple (₹20 donation)	along the Riverfront (free) • Lunch at Martin's Corner (₹900) • Explore Cabo de Rama Fort (free, great views) • Return scooter, collect luggage, head to airport/train (auto ₹400)	Scooter 350 + temple 20 + lunch 900 + auto 400 = ≈1,670
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Totals

- Accommodation (3 nights): 2,800 ₹ × 3 = 8,400
- Transport (scooter + autos): ~1,300
- Food & drinks: ~2,500 ₹ per day × 4 = 10,000
- Entry fees / donations: ~200 ₹
- Miscellaneous (snacks, water, small souvenirs): ~2,000

Grand estimate: = 22,000 ₹ – leaves ~8,000 ₹ cushion for any extra meals, a night-cap, or unexpected costs.

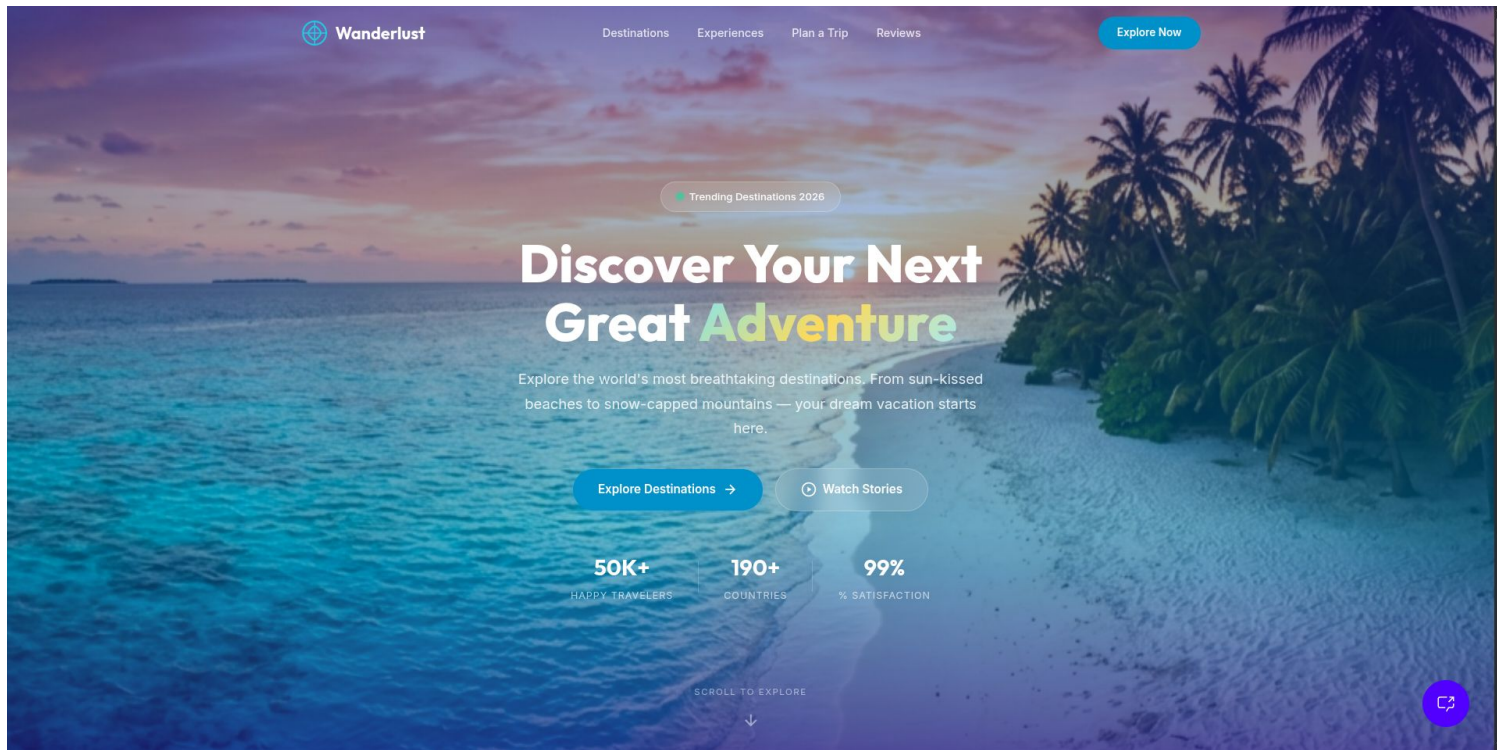
Quick Tips

- Book the hotel early – Casa De Goa often has "early-bird" discounts.
- Rent a scooter for flexibility; carry a helmet and a copy of your ID.
- Carry cash for small entry fees and temple donations (most places don't accept cards).
- Dress modestly at temples and churches (cover shoulders/knees).
- Sun protection – hat, sunscreen, and stay hydrated.

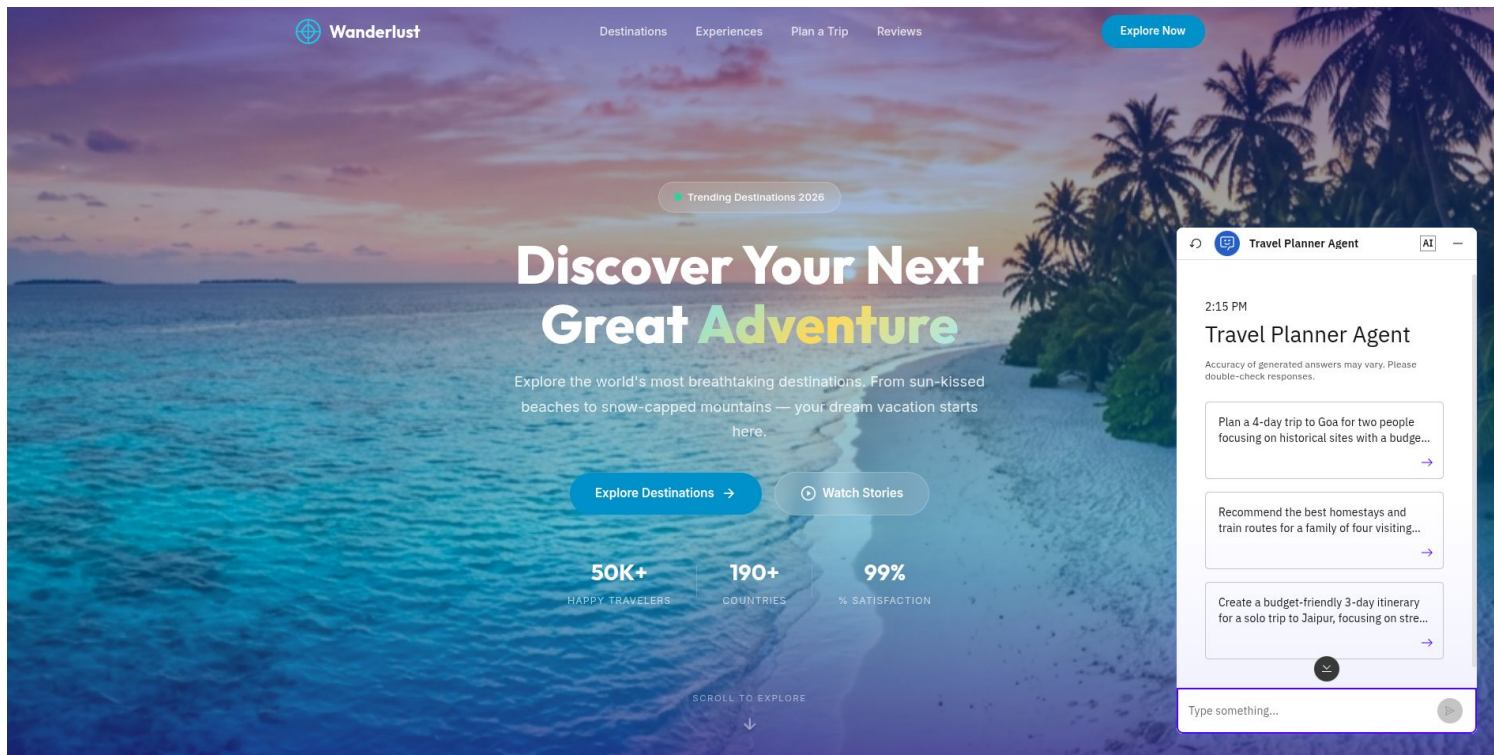
Enjoy your historic adventure in Goa! 🇮🇳 🍹 🍽️

👍 🗨️ 🔄

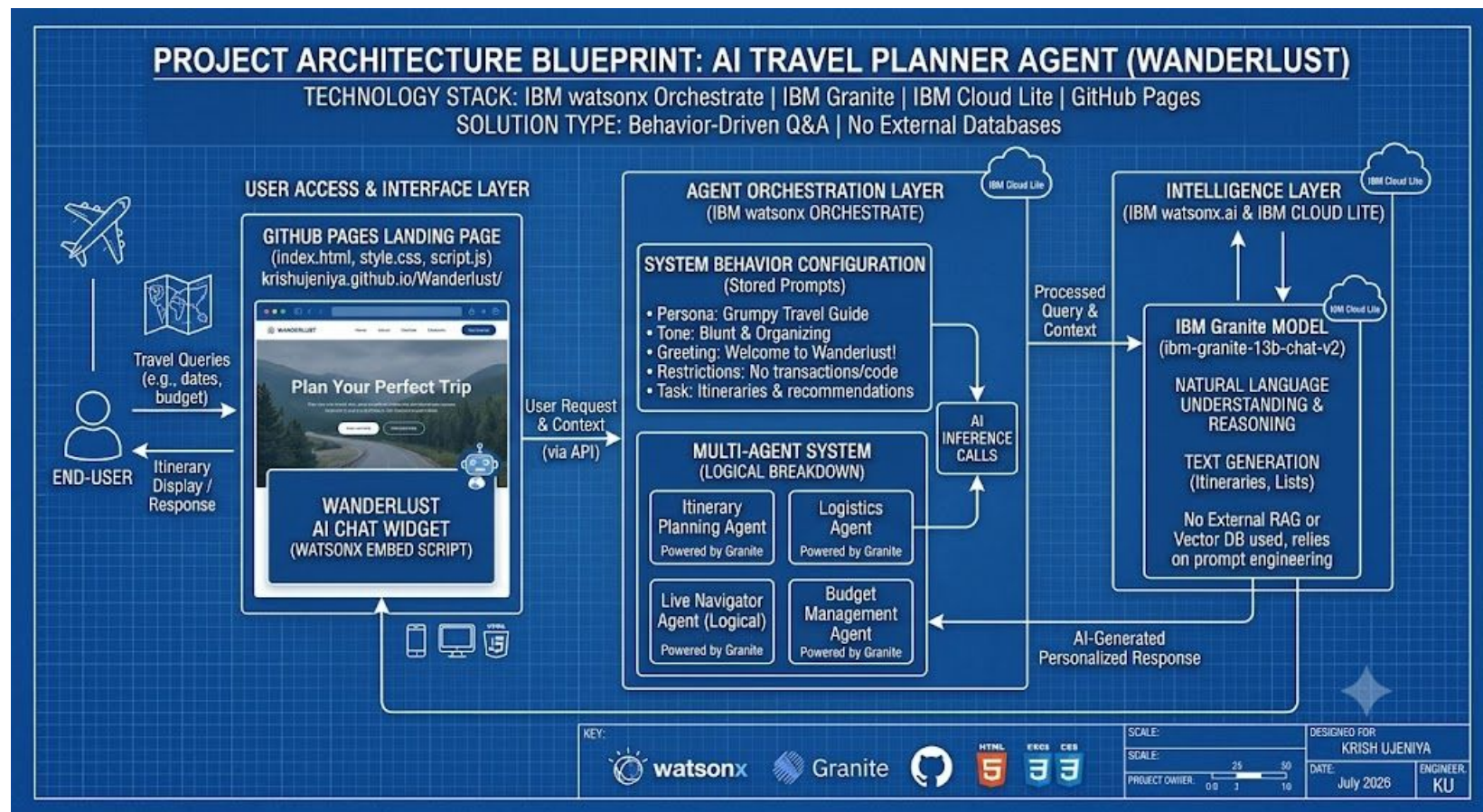
Web Integration Screenshot:



Web Integration Screenshot:



Orchestrate architecture blueprint



GitHub Link

Repo Link :

<https://github.com/krishujeniya/Wanderlust.git>

Website Link :

<https://krishujeniya.github.io/Wanderlust/>

Future Scope

1. Live Booking API Integration Right now, the agent suggests great hotels and flights based on its prompt engineering. In the future, we can connect it to live travel APIs (like Skyscanner, Amadeus, or Booking.com). This means the agent will not just suggest a flight; it will fetch the exact live ticket price and allow the user to book it directly through the chat.

2. Voice-Activated Travel Assistant Typing out long travel plans can be tedious on a phone. A major future upgrade is adding voice recognition. IBM watsonx Orchestrate is actively adding voice and phone capabilities. Users could just open the app and say, "Plan a weekend trip to Goa under 20,000 INR," and the agent will talk back with the itinerary.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Krish Ujeniya

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Getting Started with Cybersecurity



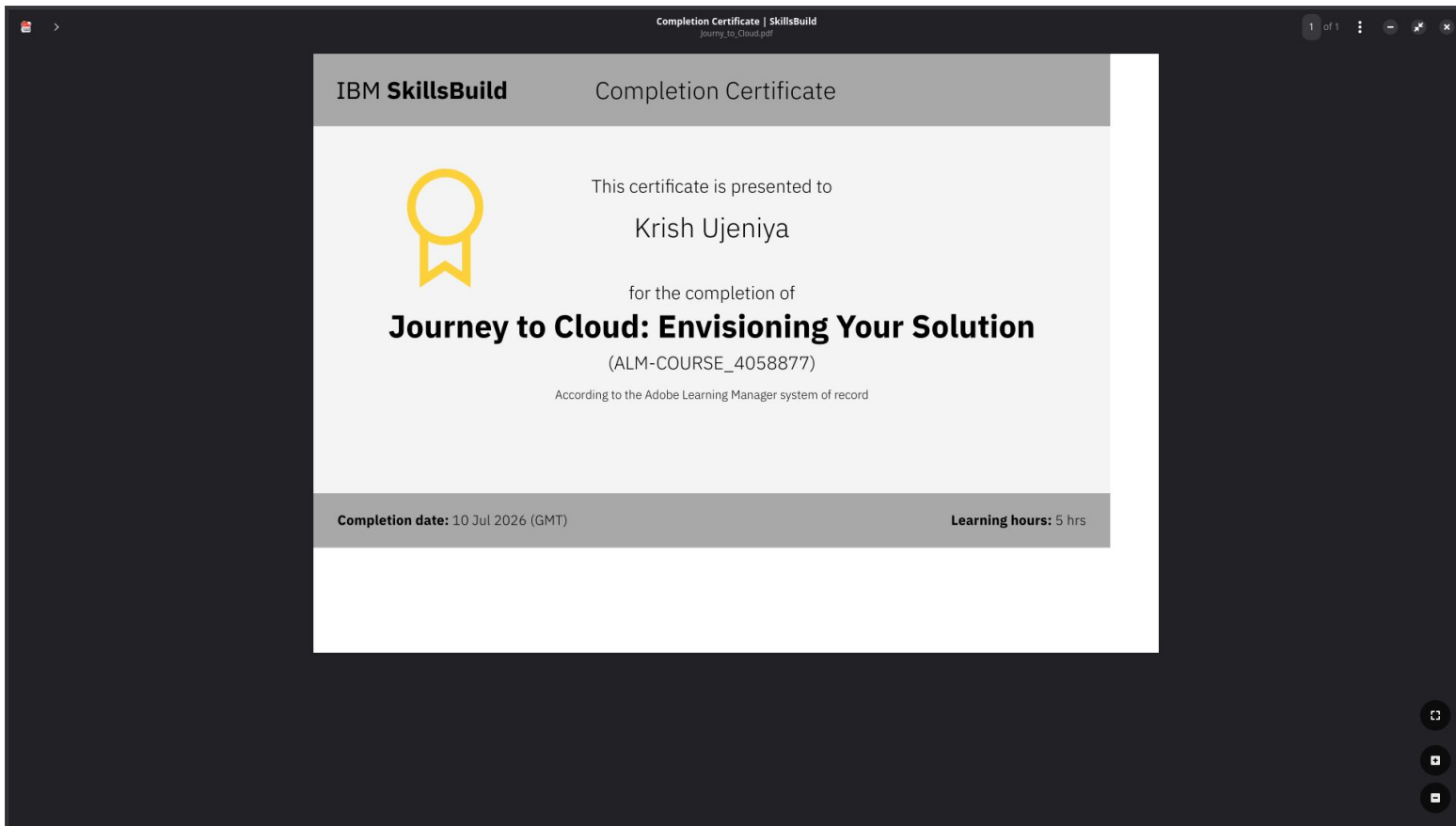
Issued on: Jul 10, 2026

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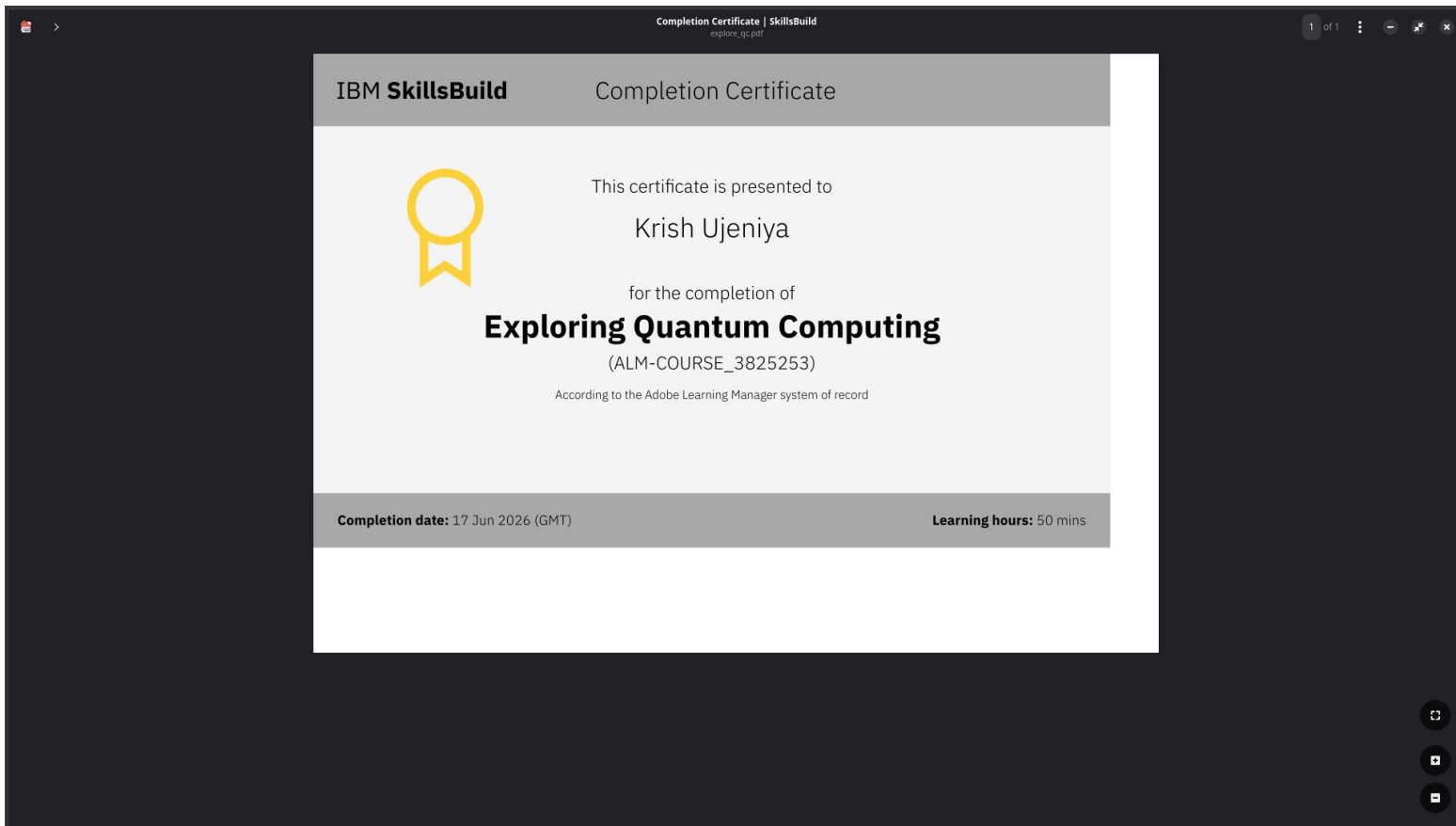
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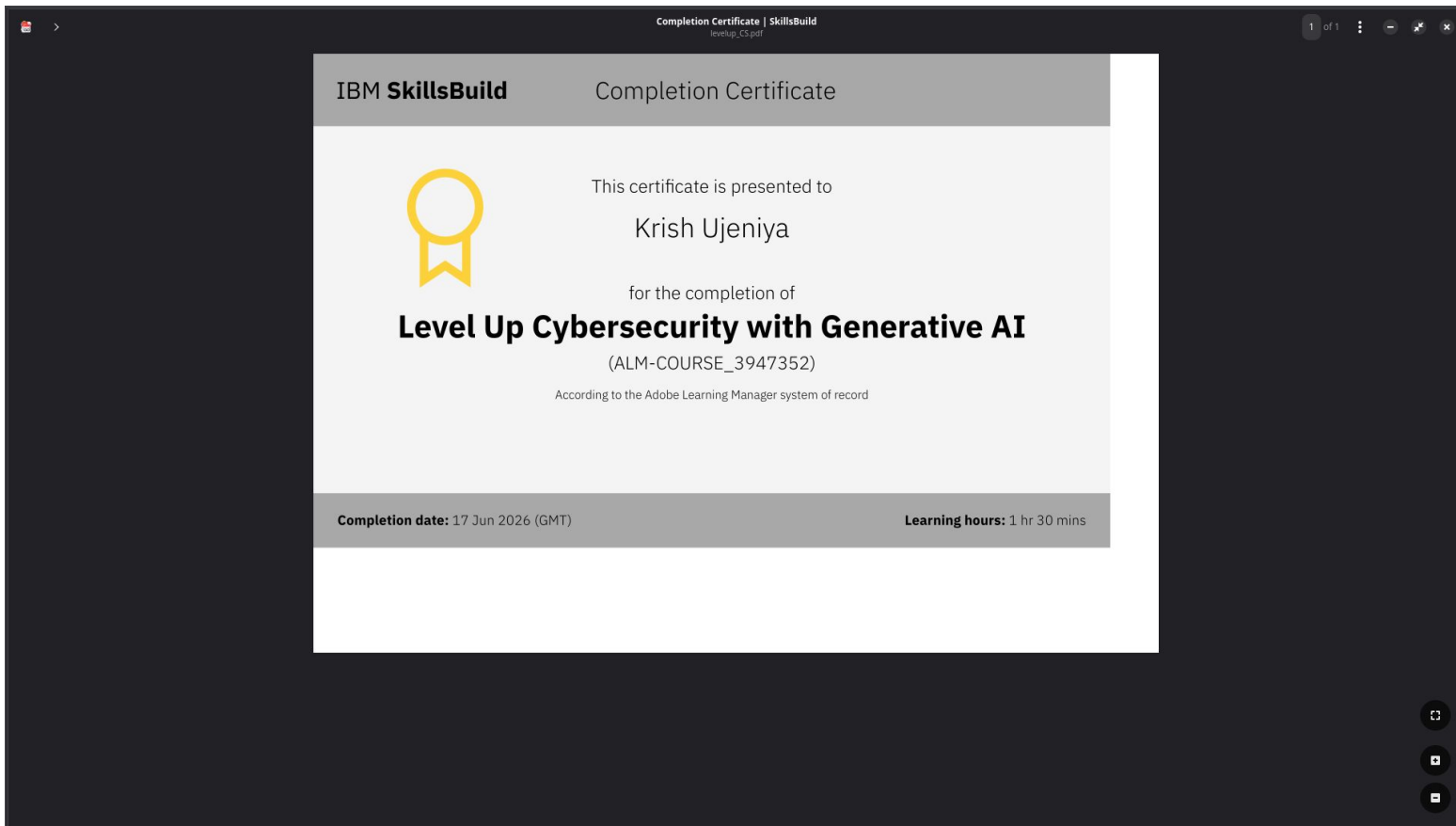
Certificates Earned:



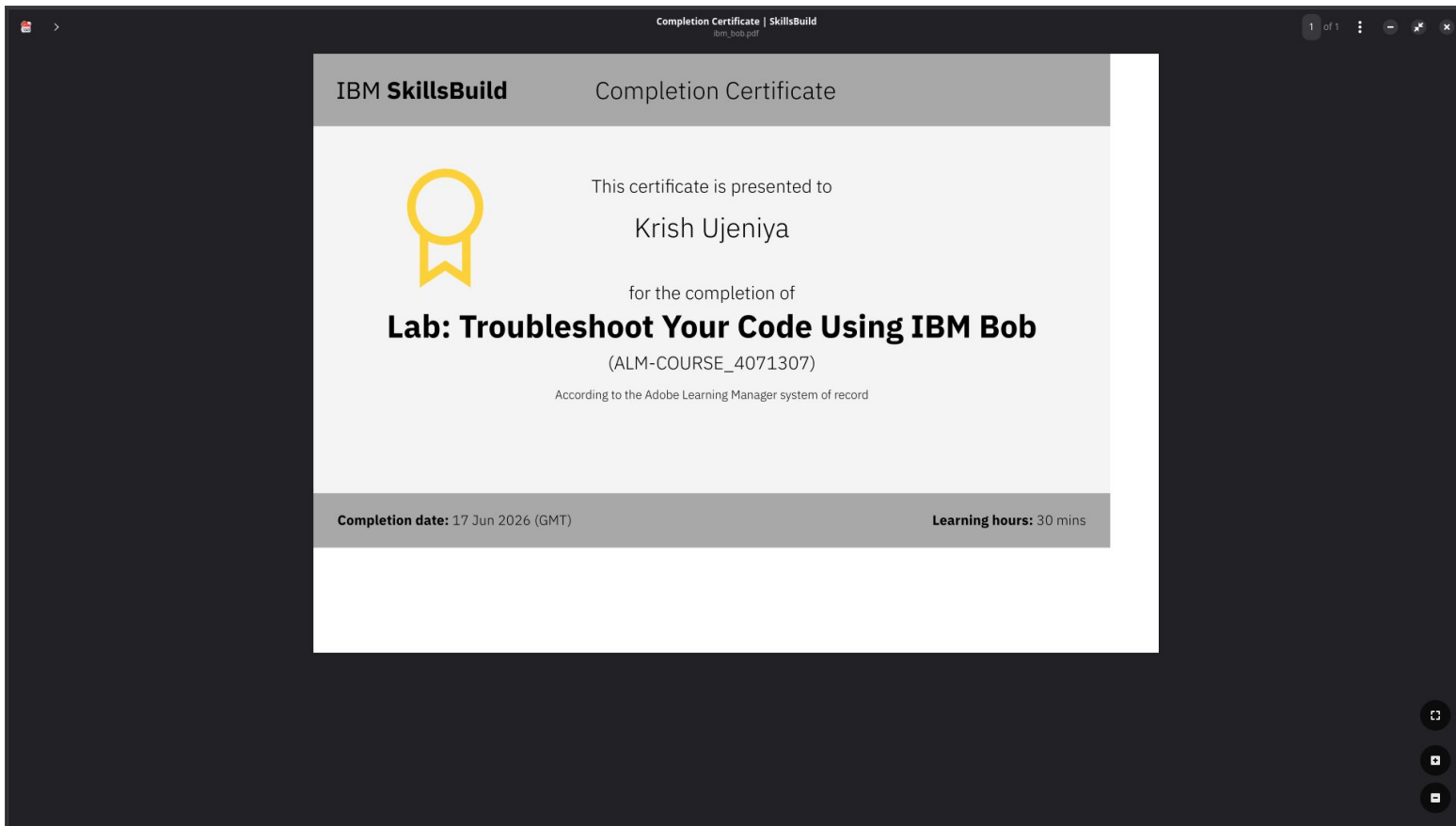
Certificates Earned:



Certificates Earned:



Certificates Earned:



Thank You!

Thank you for your time and interest.